

KATOWICE MUCHOWEC, Poland

WMO: 125600

Lat: 50.2406N

Lon: 19.0328E

Elev: 281

StdP: 98.00

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-11.7	-17.5	0.8	-14.3	-14.5	1.1	-10.8	10.3	5.0	9.1	3.9	1.5	60	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.1	20.0	28.1	19.1	26.4	18.3	21.0	27.5	20.1	26.3	19.3	24.9	3.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	14.2	23.4	18.0	13.4	22.3	17.2	12.7	21.5	61.7	27.5	58.7	26.6	55.9	25.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.2	6.3	-18.4	33.0	3.9	1.5	-21.3	34.1	-23.6	35.0	-25.8	35.8	-28.6	36.9		
(5)			-18.7	22.8	3.9	0.9	-21.5	23.5	-23.8	24.0	-26.0	24.5	-28.8	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-1.6	-0.1	3.4	8.9	13.5	16.9	18.9	18.4	13.7	8.9	4.3	-0.2	(6)
(7)		DBStd	8.32	5.25	5.14	4.38	4.27	3.77	3.51	3.11	3.29	3.45	4.15	4.31	4.86	(7)
(8)		HDD10.0	1499	360	282	207	69	11	1	0	0	7	71	175	317	(8)
(9)		HDD18.3	3606	618	516	463	283	155	68	32	41	144	291	421	575	(9)
(10)		CDD10.0	1059	0	0	3	36	119	208	274	259	117	39	4	0	(10)
(11)		CDD18.3	125	0	0	0	1	5	26	48	42	4	0	0	0	(11)
(12)		CDH23.3	1326	0	0	0	14	87	272	503	400	49	1	0	0	(12)
(13)		CDH26.7	341	0	0	0	1	10	66	146	111	7	0	0	0	(13)
(14)	Wind	WSAvg	2.7	3.2	3.1	3.1	2.7	2.4	2.4	2.3	2.2	2.3	2.5	2.8	3.2	(14)
(15)	Precipitation	PrecAvg	776	46	42	50	50	85	91	107	80	74	53	54	44	(15)
(16)		PrecMax	1106	107	104	112	109	315	187	300	166	164	109	98	133	(16)
(17)		PrecMin	584	14	12	21	0	24	35	10	13	18	6	0	14	(17)
(18)		PrecStd	120	21	22	23	24	54	38	58	35	36	31	23	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	13.1	18.2	25.2	27.9	31.2	32.5	32.7	27.4	22.5	17.3	12.1	(19)
(20)			MCWB	7.6	8.1	10.1	14.7	17.2	20.3	20.6	20.3	17.8	15.7	12.2	8.7	(20)
(21)		2%	DB	8.4	10.6	15.1	22.1	25.8	28.7	30.2	29.5	24.6	20.2	14.9	9.2	(21)
(22)			MCWB	6.3	6.9	8.9	13.1	16.7	19.5	19.9	20.0	17.2	14.7	11.0	6.9	(22)
(23)		5%	DB	6.7	8.8	12.8	19.5	23.7	26.4	28.1	27.4	22.5	17.9	12.8	7.6	(23)
(24)			MCWB	5.1	6.1	8.0	11.7	15.7	18.2	19.3	19.3	16.2	13.8	10.1	5.9	(24)
(25)		10%	DB	5.2	6.8	10.5	17.2	21.5	24.5	26.2	25.4	20.5	15.6	10.9	6.0	(25)
(26)			MCWB	3.7	4.6	6.8	10.6	14.7	17.4	18.5	18.3	15.4	12.7	8.8	4.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	8.8	11.0	15.2	18.9	21.6	22.2	21.9	19.2	16.7	12.7	9.2	(27)
(28)			MCDWB	10.0	11.5	16.2	23.3	25.8	28.3	29.2	28.3	25.1	20.9	16.0	11.4	(28)
(29)		2%	WB	6.5	7.4	9.5	13.7	17.5	20.3	21.0	20.7	18.0	15.2	11.5	7.2	(29)
(30)			MCDWB	8.0	10.0	14.0	20.4	23.6	26.6	27.5	27.3	22.5	18.9	14.3	8.9	(30)
(31)		5%	WB	5.1	6.2	8.4	12.5	16.4	19.2	20.1	19.9	17.0	14.1	10.2	5.9	(31)
(32)			MCDWB	6.5	8.5	12.1	18.1	22.1	25.0	26.2	26.0	21.3	17.3	12.4	7.2	(32)
(33)		10%	WB	3.9	4.7	7.1	11.3	15.4	18.1	19.3	19.0	16.0	12.9	8.9	4.8	(33)
(34)			MCDWB	5.1	6.6	10.1	16.2	20.5	23.2	24.7	24.3	19.8	15.5	10.6	6.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.5	6.6	8.4	10.9	11.1	10.6	10.9	11.1	9.8	8.8	6.6	5.3	(35)
(36)			MCDBR	6.6	9.4	13.1	15.3	14.9	14.1	14.6	14.2	13.7	12.6	10.5	6.6	(36)
(37)			MCWBR	5.1	6.4	7.9	7.8	7.0	6.4	6.1	6.2	7.1	7.6	7.1	5.0	(37)
(38)		5% WB	MCDBR	6.2	8.2	11.6	13.2	13.1	12.5	13.0	12.7	12.2	11.6	9.3	6.2	(38)
(39)			MCWBR	5.0	5.9	7.4	7.2	6.6	6.0	5.9	5.8	6.6	7.2	6.6	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.315	0.342	0.382	0.434	0.428	0.428	0.445	0.436	0.405	0.372	0.346	0.313	(40)	
(41)		taud	2.287	2.267	2.221	2.141	2.203	2.237	2.200	2.230	2.288	2.351	2.364	2.351	(41)	
(42)		Ebn at Noon	704	781	811	806	830	833	810	796	781	738	665	655	(42)	
(43)		Edh at Noon	70	92	115	139	136	133	136	126	107	85	66	58	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.83	1.57	2.69	4.05	4.92	5.21	5.01	4.50	3.21	1.92	0.95	0.65	(44)	
(45)		RadStd	0.15	0.26	0.41	0.51	0.60	0.54	0.62	0.39	0.46	0.27	0.13	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.45	N/A	N/A	+0.85	N/A	N/A	N/A	N/A	N/A	+31
(47) Regional (37 neighbors)		+0.51	N/A	N/A	+0.79	+0.39	+0.35	N/A	-163	+67	+26

Nomenclature: See separate page